

Revision Notes for Class 7 Science Chapter 5 – Physical and Chemical Changes

Every day, we observe different kinds of changes happening around us. Some are easily noticed, like melting of ice cubes or boiling water, while others like rusting of iron or the ripening of fruits take longer. Understanding what types of changes are happening is important for recognizing how substances behave and interact in our environment.

Physical and Chemical Changes

A change can be called physical when the original substance only changes its appearance or state. In these changes, no new substance is formed. For example, when you fold paper to make an object and then unfold it, the paper remains the same. The process of inflating and deflating a balloon, or melting ice into water, are also examples. Only the shape or state changes, not the substance itself.

Chemical changes, on the other hand, create one or more new substances. These changes are permanent and usually cannot be reversed easily. For instance, when iron rusts, it forms iron oxide, which is new. When you burn wood or cook vegetables, a chemical change happens because something new is produced that cannot be changed back into its original form easily.

Observing Changes Around Us

Let's look at some common examples from daily life:

- Melting ice cubes (Physical change)
- Chopping vegetables (Physical change)
- Boiling water (Physical change, reversible)
- Making popcorn from corn (Chemical change, irreversible)

- Burning wood (Chemical change)
- Drying wet clothes (Physical change, reversible)
- Making dough balls and rolling chapatis (Physical change)

Some changes can be noticed by our senses—sight, smell, touch, hearing, and taste—like noticing the smell of a ripening banana or watching ice melt into water.

Experiments and Activities

Many activities help us identify if a change is physical or chemical:

- Folding and crushing paper, inflating and popping a balloon, crushing chalk – these are **physical changes** as the material remains the same.
- Blowing air into tap water creates bubbles (physical), but blowing into lime water turns it milky, showing a **chemical change** as new substances like calcium carbonate are formed.
- Mixing vinegar (or lemon juice) and baking soda releases carbon dioxide (chemical change) as we can see the fizz and milky lime water test that follows.

The turning of lime water milky is commonly used to test for the presence of carbon dioxide—it's a visible result of a chemical reaction.

Chemical Changes in Everyday Life

Rusting and combustion are prime examples of chemical changes. Rusting happens when iron reacts with moisture and air, forming brownish iron oxide—something new. Combustion, like burning a candle, paper, or wood, involves reacting with oxygen and always produces new substances, heat, and light. These changes cannot be reversed easily. That's why burnt food cannot be changed back into raw form.

Experiments such as burning magnesium ribbon prove that chemical changes involve both new substance formation (white powder magnesium oxide) and release of energy (heat, light).

Combustion and Ignition Temperature

For burning (combustion) to take place, three things are needed: a combustible substance (fuel), oxygen, and heat (to reach ignition temperature). The ignition temperature is the lowest temperature required to start burning. For example, paper burns only when provided with sufficient heat (like from a matchstick or focused sunlight using a magnifying glass).

Safety tips related to fire were also discussed, such as not using synthetic materials to cover a burning person (they may melt and stick to the skin) and the importance of cutting off air supply to extinguish fire.

Some interesting facts include bioluminescence in fireflies—a chemical change that causes them to glow at night.

Physical and Chemical Changes Together

Sometimes, both types of changes occur together. For example, when a candle burns, melting and evaporation of wax are physical changes, but the burning of the vapour (producing new gases) is a chemical change. Similarly, cooking food involves irreversible chemical changes, but cutting and mixing ingredients before cooking are physical changes.

Reversible and Irreversible Changes

Some changes can be reversed, such as freezing melted ice back to solid ice or evaporating and then condensing water. These are usually physical changes. In contrast, chemical changes such as rusting, burning, and cooking are generally irreversible. Tables, like Table 5.2, help compare whether certain changes can bring back the original state or not:

S.No.	Change	Can be brought back?
1.	Melting ice cubes	Yes
2.	Chopping vegetables	No
3.	Boiling water	Yes
4.	Making popcorn from corn	No

Desirable vs Undesirable Changes

Changes can be desirable (useful), like milk turning to curd, ripening of fruits, and cooking, or undesirable like rusting of iron or spoilage of food. Some changes may be useful in one context but unwanted in another, such as decomposition (useful for compost, unwanted for stored food).

Natural Changes: Weathering and Erosion

In nature, rocks break down into smaller particles by weathering and erosion, often caused by water, wind, or even growing plant roots. This process includes both physical changes (breaking into smaller pieces) and chemical changes (formation of new substances like iron oxide). Erosion helps in forming new soils but may also cause problems like landslides.

Key Points Summarized

- Physical changes affect the appearance or physical properties but do not form new substances.
- Chemical changes form new substances, often cannot be reversed, and may release or absorb energy.
- Some changes can be both physical and chemical (like burning a candle).

- Combustion, rusting, and cooking are important chemical changes in daily life.
- Ignition temperature refers to the minimum temperature needed to start combustion.
- Some changes in nature, like weathering and erosion, happen slowly over long periods and can be both physical and chemical.

Practice and Application

To learn more, try observing changes while preparing food in the kitchen, notice which can be undone and which cannot, or perform safe experiments like mixing baking soda and vinegar. Answering the chapter-end questions can also strengthen your understanding of physical and chemical changes and their importance in our world.